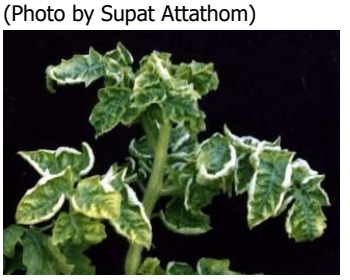
Yellow leaf curl virus management in tomatoes

Recognize the problem

Tomato yellow leaf curl virus (Virusi njano kujikunja majani, rasta, bondia in Swahili) is a disease. It can infect all growth stages but is most problematic in young stages. Infected seedlings grow slowly. The plant becomes stunted and bushy. The leaves of affected plants remain small or roll upward from the edges forming a cup-shape. Leaves become yellow between veins and at leaf edges. Less flowers and fruits grow.

Light green or mottled yellow leaves attacked by leaf curl virus. (Photo by Supat Attathom)



Background

This virus is not transmitted by soil or in the air, and not by seeds. It is transmitted by insects, mainly by whiteflies. The virus can also affect beans, red pepper, and tobacco and some weeds. Infected pepper plants will show no symptoms of virus infection but whiteflies can spread it. When plants with virus symptoms are observed in the field, consider action. Also slightly shake plants in seed beds or young plantings to check for the virus-transmitting white flies. When white flies fly off from 5 - 10 plants, they need to be controlled. If there are no virus symptoms, some whiteflies can be tolerated.

Management

Only 2 mm small white flies carry the virus. You can see the flies flying off when you shake plants. (Photo by Stefan Toepfer, CABI)



It is easier to prevent this disease than to control it:

- Use tolerant varieties (e.g. Tengeru 97); no resistant variety is available.
- Remove volunteer or leftover tomato plants and voluntary solanaceous plants, such as red pepper.

To control the disease:

- Uproot all diseased plants; bury, or put into plastic bags in the field and burn outside field.
- Viruses cannot be controlled by chemical pesticides, but the insects that spread the virus can be.

To control white flies on seedbeds or seedlings:

- Spray botanicals containing nimbecidine (Neem extract) max 3 times. Start spraying 7 days after transplanting (3 litres/ha or 60 ml/20 litres of water) (25 to 50 g neem seed cake or powder/litre water also work).
- Liquid soap spray (stir 3 tablespoons liquid soap, e.g. Cera-liquid, with 4 litres of water) is effective in the morning during wet weather. Spray every week on young plants as long as you see whiteflies.
- You may use chemical products that contain:
- Thiamethoxam; slightly hazardous (pre-harvest interval 7 days), or
- Pirimi-phosmethyl; moderately hazardous (WHO toxicity class II), max 2 sprays per season, pre-harvest interval 21 days.
- Always read and follow product label. Always wear protective clothing when mixing and spraying. Nobody should enter the sprayed field for 3 days.

When using a pesticide, always wear protective clothing and follow the instructions on the product label, such as dosage, timing of application, and pre-harvest interval.

Scientific name(s) **> Tomato Yellow Leaf Curl Virus (TYLCV)**

The recommendations in this factsheet are relevant to: Ghana, Tanzania

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